

Week 12: Final demonstration and explanation

Generative AI | MSc teaching notes and further reading | 2027

Learning target

Demonstrate the policy assistant and explain its limits with evidence.

Context and teaching plan

A fresh student question and a previously unseen failure are presented.

Contact time	Suggested use
Two hours: concepts	Case introduction, four ideas, worked example and misconception.
Two hours: practice	10 min recall, 20 min demo, 40 min guided practice, 25 min independent change, 15 min review, 10 min exit check.

Explain the key ideas

Integrated system

The student question enters retrieval, access checks, answer construction and evaluation.

Teacher question: Where does integrated system appear in the weekly case, and what could go wrong if we omit it?

Attribution

Each learner shows the part they changed and explains the choice.

Teacher question: Where does attribution appear in the weekly case, and what could go wrong if we omit it?

Fresh challenge

A new example of a practised failure tests understanding rather than memorization.

Teacher question: Where does fresh challenge appear in the weekly case, and what could go wrong if we omit it?

Handover

Another person can run, test and review the solution from written instructions.

Teacher question: Where does handover appear in the weekly case, and what could go wrong if we omit it?

Worked example and answer guide

- Input: P4: appeals must be filed within ten days.
- Change or question: Q: May I appeal on day 12?
- Supported observation: A: No; the ten-day window has passed [P4], subject to the policy version used.

Reasoning: state the given evidence first, then the calculation or rule, then the conclusion. If a value is absent, say what cannot be concluded. Keep any simulated mechanism distinct from a trained model or deployed service.

Teach the week as a story

The fresh policy question: A student appeals on day 12. P4 says appeals must be filed within ten days. The team must answer, cite, test and explain a fresh failure.

Ask for a quick prediction before each technical explanation. Return to the prediction after the example and ask what evidence changed the answer. The analogies below are memory aids, and each has a stated limit.

Integrated system: analogy, mechanism and check

Familiar picture: A museum guide checks which rooms a visitor may enter before quoting a label.

Familiar part	Technical meaning
Visitor role	access check
Exhibit label	retrieved P4
Guided answer	cited response

Where this analogy breaks: A citation needs to support the exact statement and the version in force.

Actual idea: The student question enters retrieval, access checks, answer construction and evaluation.

Work it aloud: For day 12 under the supplied ten-day P4, answer no and cite P4 with the used version.

Ask before revealing: Which three checks precede a confident reply?

Supported response: Allowed source, relevant passage and supported claim.

Look for this misconception: An answer is valid because it includes [P4].

Attribution: analogy, mechanism and check

Familiar picture: A science notebook shows a changed setting, observation and explanation.

Familiar part	Technical meaning
Setting change	your code change
Measurement	experiment table
Reason	individual explanation

Where this analogy breaks: A before/after result in one toy case cannot establish general reliability.

Actual idea: Each learner shows the part they changed and explains the choice.

Work it aloud: Show baseline, chosen setting, case counts and one error that still remains.

Ask before revealing: What proves the work is yours and reproducible?

Supported response: The changed code, settings, inputs and explanation.

Look for this misconception: A polished demonstration alone proves understanding.

Fresh challenge: analogy, mechanism and check

Familiar picture: A mechanic diagnoses a new symptom by testing the failing part.

Familiar part	Technical meaning
Symptom	new failure
Component	likely layer
Check	targeted test

Where this analogy breaks: Some failures cross layers; record uncertainty and test competing causes.

Actual idea: A new example of a practised failure tests understanding rather than memorization.

Work it aloud: If P4 is retrieved but the answer says 12 days is allowed, test claim support before changing search.

Ask before revealing: Which layer deserves the first test here?

Supported response: Answer support using the retrieved P4 passage.

Look for this misconception: Every wrong answer is a retrieval failure.

Handover: analogy, mechanism and check

Familiar picture: A relief teacher can run a lesson from a clear folder and known limits.

Familiar part	Technical meaning
Folder	setup guide
Known gaps	limitations
Trial run	repeatable handover

Where this analogy breaks: Documentation can go stale; date versions and rerun the checks.

Actual idea: Another person can run, test and review the solution from written instructions.

Work it aloud: Give installation steps, source versions, test commands, role rules and rollback instructions.

Ask before revealing: What can the next maintainer reproduce?

Supported response: The setup, evaluation and known-failure checks from the handover.

Look for this misconception: The notebook is self-explanatory without instructions.

New situation for discussion

A new policy edition changes the appeal window. What records, index and tests need updating?

Memory cue: Show the question-to-evidence path, your measured change and a repeatable handover.

Guided laboratory

Present a self-contained end-to-end policy assistant.

1. Run an answerable student policy question
2. Run a no-answer question
3. Demonstrate blocked staff-only access
4. Explain one failure and propose a new test

Individual change: Use a new policy version and explain which index and tests must change. Expected evidence: Demo, test table, known limits and individual explanation.

Checks and common error

Measure: Task success, citation support, no-answer accuracy and access gate. Record the input, setting, source version, result and the relevant numerator and denominator.

Plausible incorrect claim: A polished demonstration without tests establishes that the system is ready.

Correction: Show the test cases, negative cases, access result and remaining limitations.

If learners disagree, reproduce the same case and compare the underlying source or expected answer. Record unresolved cases for instructor review.

Further discussion

- How could the conclusion change if the source or input changes?
- Which part of the result is observed and which part is an assumption?
- What simple test would detect the failure described above?

References and optional extension

Core readings are official vendor documentation or original research. Read the relevant section; an account or paid API is unnecessary for this notebook. Page details may change before 2027.

Microsoft, Retrieval augmented generation.

<https://learn.microsoft.com/en-us/azure/foundry/concepts/retrieval-augmented-generation>

OpenAI, Evaluation best practices. <https://developers.openai.com/api/docs/guides/evaluation-best-practices>

Anthropic, Building effective agents. <https://www.anthropic.com/engineering/building-effective-agents>

Optional extension: replace one classroom proxy with a locally approved model or service, keep the original baseline, repeat the same checks, and state the additional privacy and cost assumptions.

Curriculum scope: the supplied 12-week teaching sequence, recorded in docs/SYLLABUS_ALIGNMENT.md. The original module descriptor is retained in sources/originals/07-Descriptor-Generative-AI-new-.docx. Supporting project notes: Deep-research.txt, RAG.txt, AI-Agents.txt, MCP-&-Vector-Database.txt, Multi-Agent-AI-System.txt, Eval-&-Fine-tune-&-Loop-Engineering.txt. These are planning sources, not empirical performance evidence.

Student revision and exam preparation

Outcomes: LO1–LO5 | 2 hours concepts + 2 hours guided practice

The idea in plain language

A final demonstration must show an answerable case, an unsupported case, a denied-access case and recovery from a failure. Each learner explains their own change. Compare against week 1 only on matched questions with consistent scoring.

A worked example

Run the same two travel and appeal questions on baseline and final versions. Count supported correct answers under the same rubric. Explain changed settings and unresolved failures; do not compare a ten-question baseline percentage with a different final dataset as if they were controlled.

Practice before reading the answers

1. What makes a fair baseline comparison? (4 suggested practice marks)
2. Name the four demonstration cases. (4 suggested practice marks)
3. What should an individual explanation include? (4 suggested practice marks)

Draw the main diagram from memory, explain each arrow, then check it against the slides. Answer the three questions without notes before reading the model answers.

Model answers and marking guidance

1. Same task, matched cases, source version and scoring rubric, with changes and variability recorded.
2. Answerable, unsupported, access denied, and failure recovery.
3. The learner's own change, why they made it, test evidence and a remaining limitation.

For each four-mark practice answer: two marks for the correct idea, one for a relevant example or calculation, one for an appropriate explanation or limitation. These suggested practice marks do not change the module assessment scheme.

Official reading and video support

M1 Microsoft Learn: Introduction to retrieval-augmented generation concepts

<https://learn.microsoft.com/en-us/training/modules/rag-fundamentals/>

Study focus: Preparation, retrieval, generation and evaluation.

O1 OpenAI: Evaluation best practices

<https://developers.openai.com/api/docs/guides/evaluation-best-practices>

Study focus: Task-specific testing, repeated outputs and separate evaluation cases.

N1 NVIDIA: Generative AI and LLM Learning Paths

<https://www.nvidia.com/en-us/learn/learning-path/generative-ai-llm/>

Study focus: Optional deeper RAG, diffusion and deployment training; some courses require payment.

Selected source concepts inform the lesson. Examples, figures, questions and solutions are original classroom material; no exam questions are claimed to be official.